Bayesian Estimation of Nonlinear DSGE Models with Neural-Network Surrogate Solvers

A Stochastic Extended Path to HMC pipeline

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Views expressed are my own and do not necessarily represent the IMF.

Main Takeaway First

Estimate with a switching solution-order likelihood.

What I do

I start each likelihood evaluation from ROM1, switch in the learned nonlinear correction only on audited support, and keep direct SEP as the teacher and audit.

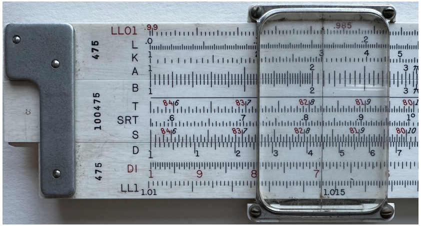
Why it matters

This puts the order of approximation inside the likelihood: cheap where the model is locally linear, richer where the posterior asks for nonlinear dynamics.

The contribution is computational and econometric: the sampler sees one differentiable target, while the solution rule switches between ROM1, ROM1+NN, and direct SEP audit.

The Intuition: Precompute the Hard Object

A surrogate is a modern lookup table for a hard economic function.



Logs and products were once handled by tables and interpolation.

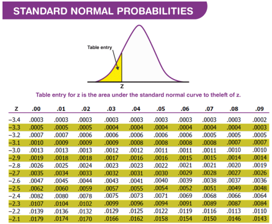

STANDARD NORMAL PROBABILITIES

Table entry

z

Table entry for z is the area under the standard normal curve to theleft of z.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
−3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
−3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
−3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
−3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
−2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
−2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
−2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
−2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
−2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
−2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
−2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
−2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
−2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143

Hard functions were made operational by precomputation.

I use the same idea for a nonlinear DSGE transition map:

$$(s_t, \varepsilon_{t+1}, \theta) \mapsto s_{t+1}^{\mathrm{SEP}}.$$

Images adapted from the previous Nonlinear DSGE presentation. Slide-rule source: Arnold Reinhold, Wikimedia Commons. Z-score table source: BYJU'S.

The Hook

Nonlinear DSGE models are economically natural and computationally awkward.

Economists care about nonlinear states.

- Effective lower bounds.
- Occasionally binding constraints.
- Large shocks and crisis periods.
- Curvature in pricing, wages, investment, and utilization.

But estimation usually retreats to linear tools.

- Perturbation is fast.
- Kalman filtering is stable.
- The full nonlinear likelihood is too expensive.
- Particle filters can be fragile in high dimension.

The question is not whether nonlinearities matter. The question is whether we can estimate them at scale.

Question

How can we estimate nonlinear DSGE models without solving the model at every posterior draw?

The empirical problem

Medium-scale DSGE models are useful because they connect mechanisms, shocks, and observables. The same structure makes nonlinear estimation expensive.

What we want

- Structural parameters with posterior uncertainty.
- Likelihood comparisons across nonlinear mechanisms.
- Diagnostics that say when linearization is misleading.

What blocks us

- Repeated nonlinear solves.
- Numerical failures away from the mode.
- Slow or noisy likelihood evaluations.

Where This Fits

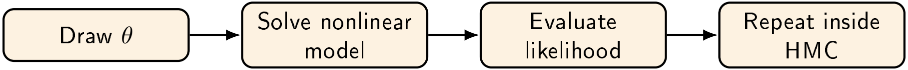
Approach	Main idea	Limitation for this talk
Particle-filter likelihoods	Integrate nonlinear latent states	Accurate; costly and noisy in large systems
Projection / OBC filtering	Approximate policy functions and split regimes	Stable near constraints; still costly in estimation
Global NN solvers / MKR	Amortize nonlinear objects over states and parameters	Need likelihood audits and support checks
This paper	**Switching ROM1/NN likelihood**	**Reported support and approximation error**

The gain is not "NN instead of economics." The gain is **ROM1 first, NN when audited, SEP as benchmark**.

Closest strands: *particle-filter likelihoods; projection and piecewise OBC methods such as Gust et al.; and neural-network global solvers such as Kase, Melosi, and Rottner (MKR).*

The Computational Problem

The nonlinear solver sits in the wrong loop.



Repeated many times inside HMC.

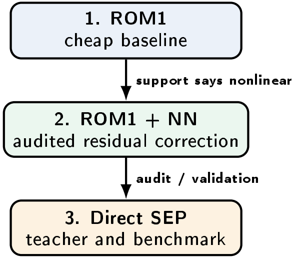
The nested-loop bottleneck

$$\theta \longmapsto g_\theta^{\mathrm{SEP}} \longmapsto p(y_{1:T} \mid \theta) \longmapsto \text{MCMC proposal}$$

The expensive object is recomputed exactly where the sampler needs speed.

The Solution in One Picture

The likelihood switches across a solution ladder, not across models.



At each likelihood date

$$\widehat{g}_t = g_\theta^{\mathrm{ROM1}} + \omega(d_t)\, r_\phi(s_t, \varepsilon_{t+1}, \theta).$$

- d_t summarizes support, inversion, and regime diagnostics.
- $\omega(d_t) \approx 0$: ROM1 dominates.
- $\omega(d_t) \approx 1$: nonlinear residual is active.
- Outside audited support: flag the draw or benchmark with direct SEP.

The object is a switching solution-order likelihood. The posterior tells us when first-order dynamics suffice and when nonlinear solution information matters.

What Exactly Is Approximated?

The neural net approximates the solved transition map, not the model equations.

Equilibrium discipline

$$F(s_{t-1}, s_t, s_{t+1}, \varepsilon_t; \theta) = 0$$

is imposed by SEP offline, including expectations and constraints.

Training object

$$g_\theta^{\mathrm{SEP}}(s_t, \varepsilon_{t+1}) = s_{t+1}^{\mathrm{SEP}}.$$

Residual target

$$r_\phi(s, \varepsilon, \theta) \approx g_\theta^{\mathrm{SEP}}(s, \varepsilon) - g_\theta^{\mathrm{ROM1}}(s, \varepsilon).$$

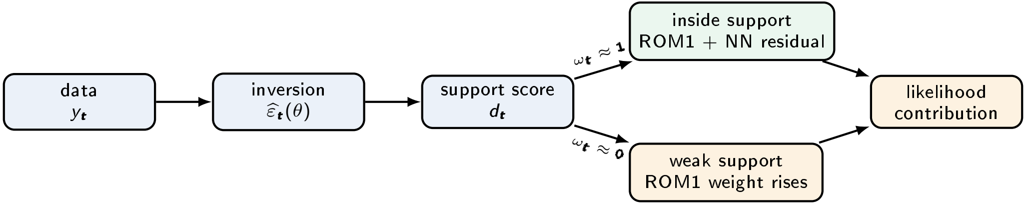
Online map

$$\widehat{g}_{\theta,\phi} = g_\theta^{\mathrm{ROM1}} + \omega(s, \varepsilon, \theta) r_\phi.$$

This distinction matters for econometrics: the approximation is a controlled change to the likelihood, not a new behavioral model.

How the Switching Rule Operates

The gate chooses the solution order period by period.

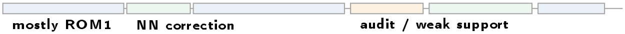


Switching transition map

$$\widehat{g}_t = g_\theta^{\mathrm{ROM1}} + \omega_t r_\phi, \qquad 0 \le \omega_t \le 1.$$

What the switch reports

- Where ROM1 is enough.
- Where the NN correction is active.
- Which periods need direct SEP audit.



Step 0: Choose the Estimation Target

Be explicit about the object before approximating it.

$$s_{t+1} = g_\theta(s_t, \varepsilon_{t+1}), \qquad y_t = h(s_t; \theta) + \eta_t, \qquad \varepsilon_t \sim \mathcal{N}(0, I), \quad \eta_t \sim \mathcal{N}(0, \Sigma_y).$$

Objects

- Structural parameters θ.
- State vector s_t.
- Structural shocks ε_t.
- Observables y_t.

Choice to make explicit

- Direct measurement-error likelihood or inversion/profile criterion.
- Which parameters vary in training and HMC.
- Parameter and shock support.
- What is benchmarked with direct SEP.

The approximation must be judged relative to this target, not relative to a generic forecast exercise.

Step 1: Solve the Nonlinear Model Offline

Use SEP as an offline teacher.

For sampled triples (s, ε, θ), compute the nonlinear transition

$$s^{+}_{\mathrm{SEP}} = g_{\theta}^{\mathrm{SEP}}(s, \varepsilon).$$

$$\mathbb{E}_t m(s_{t+1}) \approx \sum_{j=1}^{J} w_j \, m\big(g_{\theta}^{\mathrm{SEP}}(s_t, \varepsilon_{j,t+1})\big) .$$

Why SEP?

- Keeps the nonlinear equations.
- Handles occasionally binding constraints.
- Gives direct objects for validation.

What must be reported

- Solver tolerances.
- Failed and successful points.
- Shock and parameter support.
- Initialization strategy.

Step 2: Compute the Linear Baseline at the Same Points

Use the linear solution as the baseline, not as the answer.

At the same inputs, compute

$$s^+_{\mathrm{ROM1}} = g_\theta^{\mathrm{ROM1}}(s, \varepsilon).$$

Reason

The linear solution is cheap and often accurate near the steady state. The NN should not waste capacity relearning what ROM1 already does well.

$$\text{Nonlinear object} = \text{linear baseline} + \text{nonlinear correction}.$$

Step 3: Train the Residual Surrogate

Train the nonlinear correction.

Train r_ϕ on the residual

$$r_\phi(s, \varepsilon, \theta) \approx g_\theta^{\mathrm{SEP}}(s, \varepsilon) - g_\theta^{\mathrm{ROM1}}(s, \varepsilon).$$

Why residual learning

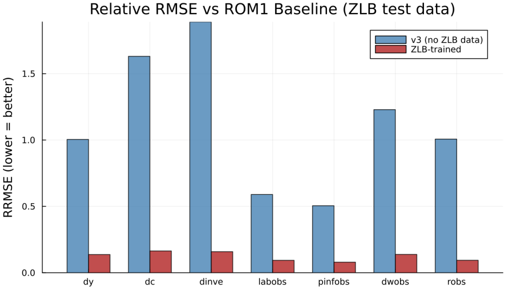
- Smaller target.
- Easier extrapolation around the linear region.
- Directly measures what linearization misses.

Resulting transition

$$g_\phi(s, \varepsilon, \theta) = g_\theta^{\mathrm{ROM1}}(s, \varepsilon) + r_\phi(s, \varepsilon, \theta).$$

Step 4: Validate the Surrogate Before Estimation

Do not estimate before the surrogate passes basic audits.



Checks

- Held-out direct SEP points.
- RMSE and relative RMSE by observable.
- Regime-specific accuracy: binding, non-binding, near-boundary.
- Finite likelihood and finite-gradient checks.

Rule

The surrogate is only credible on the support where it has been audited.

Step 5: Build the Likelihood

The likelihood determines what the validation has to prove.

Direct validation target

$$s_{t+1}^{m} = g_{\theta}^{m}(s_t^{m}, \varepsilon_{t+1}), \quad m \in \{\mathrm{SEP}, \phi\}.$$

$$\ell_m(\theta, \varepsilon_{1:T}, s_0) = \sum_{t=1}^{T} \log \mathcal{N}(y_t; h(s_t^{m}; \theta), \Sigma_y)\,.$$

This is the clean synthetic-data benchmark: direct SEP-HMC versus surrogate-HMC.

Medium-scale profiled target

$$\widehat{\varepsilon}_{t+1}(\theta) = I_{\theta}(y_{t+1}, s_t),$$

$$\widehat{s}_{t+1} = g_{\theta}^{\mathrm{ROM1}}(s_t, \widehat{\varepsilon}_{t+1}) + \omega_t r_{\phi}(s_t, \widehat{\varepsilon}_{t+1}, \theta).$$

Here inversion fixes shocks period by period; the gate reports when the criterion relies on the nonlinear correction.

Step 6: Sample Parameters with HMC

The differentiable surrogate is what makes HMC practical.

$$q = (\theta, s_0, \varepsilon_{1:T}), \qquad \log \pi_m(q \mid y_{1:T}) = \log p(\theta) + \log p(s_0 \mid \theta) + \sum_{t=1}^{T} \log p(\varepsilon_t) + \ell_m(q).$$

What is sampled

- Structural parameters θ.
- Initial condition or state objects.
- Latent structural shocks $\varepsilon_{1:T}$.
- Same prior and transforms in direct and surrogate runs.

What HMC checks

- The recursion is differentiable.
- Gradients are available through $\widehat{g}_t$.
- Divergences, energy behavior, and mixing test local geometry.
- Matched settings make direct and surrogate posteriors comparable.

The validation standard is posterior agreement within Monte Carlo error, not only a low one-step prediction error.

Step 7: Audit the Whole Pipeline

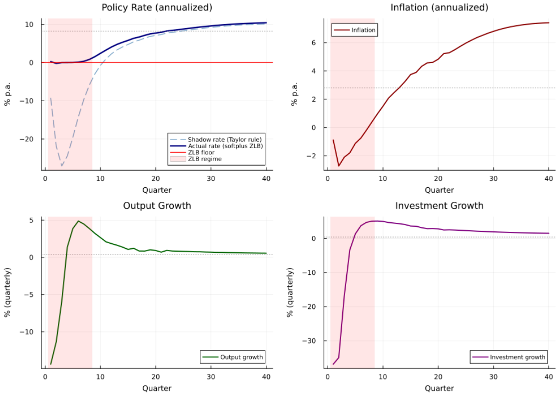
A surrogate paper lives or dies by its audit trail.

Object	Question	Failure mode
SEP support	Did the solver cover the region?	Hidden selection
Surrogate fit	Is the map accurate where used?	Biased likelihood
Gradients	Are target derivatives finite?	Bad HMC geometry
HMC chains	Do chains mix and agree?	Unreliable posterior
Direct audits	Does SEP confirm key cells?	Unsupported extrapolation

The replication package should make these checks mechanical, not narrative.

Validation: A Hard Nonlinearity

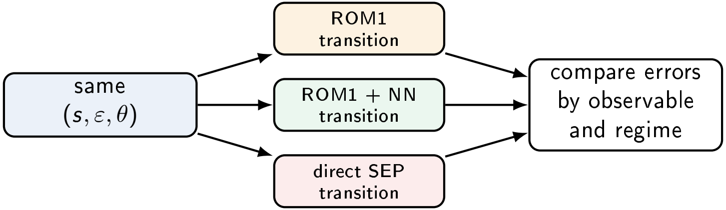
The validation needs a region where linearization actually fails.



Same shocks, nonlinear occasionally-binding model versus linearized model.

Validation: Same Inputs, Three Objects

A convincing validation holds the shock, state, and parameter vector fixed.



Accuracy question

$$\|g^{\mathrm{SEP}} - [g^{\mathrm{ROM1}} + r_\phi]\| \ll \|g^{\mathrm{SEP}} - g^{\mathrm{ROM1}}\|.$$

Econometric question

$$p_\phi(\theta \mid y_{1:T}) \approx p_{\mathrm{SEP}}(\theta \mid y_{1:T}).$$

This is why I report solver support, held-out residuals, finite gradients, HMC diagnostics, and direct SEP posterior comparisons separately.

Validation: What the Small Model Shows

The small model is a controlled benchmark for the likelihood pipeline.

Design

- Nonlinear New Keynesian prototype.
- Synthetic data generated by the nonlinear model.
- Selected shock-scale parameters estimated.
- Direct SEP-HMC compared to surrogate-HMC.

Interpretation

- The pipeline can recover parameters in a controlled nonlinear setting.
- The validation is local to the model and support.
- It is a feasibility proof, not a universal theorem.

Preliminary Application: Smets–Wouters

The medium-scale model is where the computational problem becomes real.

Empirical laboratory

Smets–Wouters style model with investment adjustment costs, variable utilization, Kimball pricing and wages, and occasionally binding constraints.

Why this is useful

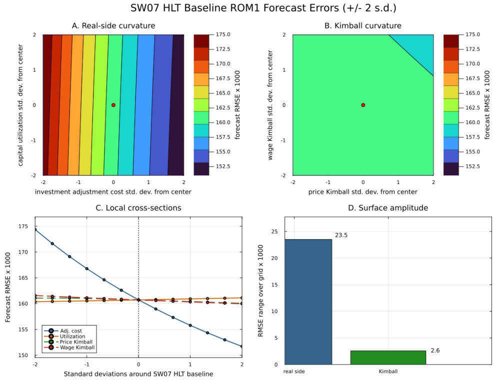
- Standard empirical DSGE laboratory.
- Rich enough to stress nonlinear mechanisms.
- Economically interpretable diagnostic shifts.

Why this is hard

- Higher-dimensional state.
- Fragile nonlinear solves.
- Strong need for out-of-support diagnostics.

Preliminary HLT Evidence

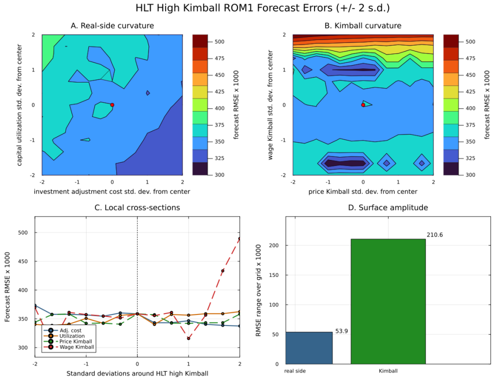
Near the posterior region, the largest local curvature is on the real side.



Preliminary diagnostic evidence: investment and utilization curvature move the SEP–ROM1 forecast error more than Kimball curvature around this posterior-region neighborhood.

Preliminary Stress Test: Kimball Can Matter Away From the Posterior Region

Kimball curvature can dominate after moving to a deliberately high-curvature neighborhood.



What changes

If the calibration is deliberately moved to a high price/wage-curvature neighborhood, Kimball curvature can dominate the local forecast-error surface.

What this means

The investment-channel diagnostic is a posterior-region finding, not a global claim about every DSGE calibration.

Computational Accounting

The computational gain is a switched target with amortized nonlinear solves.

Task	Solution order	Role
SEP solves	benchmark order	expensive teacher
ROM1 solution	first order	cheap default
ROM1 + NN residual	switched-in order	nonlinear correction
Gate/support rule	estimator	chooses order by date
HMC	posterior sampler	explores switched target

The win is not only offline versus online.
The win is that the estimator changes solution order only where audited.

Preliminary Evidence: What the Diagnostics Say

The evidence is preliminary, but the diagnostics are informative.

Method diagnostics

- Corrected unit-shock HLT MVP: 3,200 direct SEP training points.
- ROM1-residual NN cuts observable residual RMSE by 87–94 percent.
- Small-model ELB/OBC matched HMC diagnostics.
- Reduced HLT bridge grid: direct SEP, ROM1, and surrogate surfaces.
- Full-NN and linear+gate MVP HMC smokes run without divergences.

Interpretation

- The switching likelihood is operational in controlled settings.
- The medium-scale evidence should be read as diagnostic, not final.
- The full 18-parameter production posterior is still being extended.
- The relevant question is posterior-region support, not average fit.

Closing Takeaways

The method is promising because the approximation is explicit and auditable.

1. The computational obstacle is clear: global nonlinear solves are too expensive for the inner likelihood loop.
2. The proposed solution is simple: use ROM1 first, switch in the learned SEP residual where audited, and keep direct SEP as the benchmark.
3. The small nonlinear validation checks posterior agreement, not only fit.
4. The preliminary Smets–Wouters evidence is useful because it is tied to solver support, residual errors, gradients, and HMC diagnostics.

For this audience, the main object is not a black-box NN. It is a likelihood whose solution order is observed, gated, and audited.

Thank you

Questions and comments welcome.

Backup: Relation to Gust et al. and MKR

Approach	What is approximated?	How inference is done
Gust et al. style OBC estimation	Policy functions with fixed projection basis and regime handling	Particle-filter likelihood and particle-filter MCMC
Kase–Melosi–Rottner	Neural-network objects over states and parameters	Fast estimation after offline amortization
This deck	**SEP transition residual over ROM1**	**Differentiable switched likelihood for HMC**

The shared idea is amortization. The distinction is the likelihood object: I train a ROM1 residual against direct SEP and audit the posterior target explicitly.

Backup: Runtime Budget

On this workstation, budget about one week for the production run.

Component	Measured anchor	Production budget
Base SEP data	1,664 corrected unit-shock samples in 2.8 hours	2–4 days for 125 parameter blocks at longer sample and horizon
ZLB SEP data	1,536 binding samples in 6.3 hours	2–5 days to obtain about 5,000 accepted binding points
Combine and train	3,200 points trained quickly; NN lowers ROM1 residual RMSE by 87–94 percent	Less than 2 hours unless the dataset is much larger
Surrogate HMC	100 warmup + 100 draws in 25 minutes at tree depth 5	3–8 hours per 1,500-iteration chain

If base and ZLB jobs run in parallel, the end-to-end wall-clock estimate is **5–8 days**, including training and matched HMC checks.

Backup: Target and Surrogate Error

$$e_\phi(s,\varepsilon,\theta) = g_\theta^{\mathrm{SEP}}(s,\varepsilon) - \left[g_\theta^{\mathrm{ROM1}}(s,\varepsilon) + r_\phi(s,\varepsilon,\theta)\right].$$

Validation quantity

$$\mathrm{RMSE}_{\mathrm{val}} = \left(\frac{1}{N}\sum_{i=1}^{N} \|e_\phi(s_i,\varepsilon_i,\theta_i)\|^2\right)^{1/2}.$$

The same held-out inputs are evaluated under direct SEP, ROM1, and the surrogate.

Backup: Matched HMC Validation Design

- Same synthetic data.
- Same prior and parameter transforms.
- Same initial values and random seeds.
- Same NUTS settings: warmup, draws, target acceptance, tree depth.
- Compare posterior means, MCSEs, and 90 percent intervals.

Pass criterion

The surrogate posterior should overlap the direct SEP posterior within Monte Carlo uncertainty, with no systematic directional distortion.

Backup: Why SEP?

- Keeps the original nonlinear equations.
- Handles occasionally binding constraints directly.
- Produces a simulation object that can be compared with perturbation.
- Expensive, but parallel over training points.

Tradeoff

SEP is a good offline teacher, but a poor online likelihood engine.

Backup: Posterior Interpretation

- The small-model validation targets direct likelihood comparisons.
- The medium-scale empirical exercise uses an inversion/profiled criterion.
- Surrogate HMC results should be read as criterion-based posterior diagnostics until a full direct-SEP HMC benchmark is feasible.

The method is promising because the validation pieces work, not because every medium-scale empirical caveat is already solved.